CIS 5500 Database and Information Systems
Final Project Proposal - Group 6
Flight Price and Reliability Analysis

Group Member	Email	GitHub Username
Sabina Lu	sabinalu@engineering.upenn.edu	sabina-lu
Peggy Che	yidanche@engineering.upenn.edu	yc28C
Judy Li	li317@engineering.upenn.edu	judylishihan
Crystal Wang	ihw@engineering.upenn.edu	IhsuanWang

Application / Website Idea:

Our project aims to develop a web-based platform that helps travelers analyze historical flight prices and reliability across different airlines and routes. By integrating flight pricing data with flight performance and weather data, the website allows users to explore ticket price trends, flight delays, cancellations, and weather-related disruptions.

Users will be able to filter data by origin, destination, airline, and historical dates. The website will provide analysis of ticket prices across different booking windows, airline price comparisons, flight delay and cancellation rates, weather-related travel risks, and historical airline performance.

Datasets:

Dataset 1. Flight Prices

a. Description: This dataset contains historical flight ticket prices collected from Expedia between April and October 2022. It includes flight search dates, departure dates, airlines, routes, ticket prices, travel duration, and other flight information.

b. Source: https://www.kaggle.com/datasets/dilwong/flightprices

d. (i) Relevant size: 31.09 GB, 27 cols, 82,138,753 rows

(ii) Summary statistics: Due to the large dataset size, we calculated summary statistics using a subset of 20,000,000 rows. The **days_before_departure** attribute was derived to measure the number of days between the flight search date and the scheduled departure date.

Attributes	Mean	Std	Min	Median	Max
totalFare	383.89	211.99	23.97	351.6	8260.61
totalTravelDistance	1566.35	842.26	97.00	1392.00	4430.00
days_before_departure	27.43	17.25	1.00	26.00	42.00

Flight Type Distribution: non-stop flights: 29.29%, connecting flights: 70.71%

Dataset 2. US Flight Delays 2025 (BTS On-Time Performance)

a. Description: This dataset contains 7,001,619 scheduled U.S. domestic flights from January to December 2025, sourced from the Bureau of Transportation Statistics (BTS). It includes flight dates, airlines, routes, scheduled and actual departure/arrival times, delays, cancellations, diversions, flight distances, and delay causes (carrier, weather, airspace system, security, and late aircraft).

b. Source:
https://www.kaggle.com/datasets/a7madmostafa/us-flight-delays-2025-bts-on-time-performance

d. (i) Relevant Size Statistics: 1.41 GB, 7,001,619 rows, 40 attributes, covering 352 distinct airports from January to December 2025.

(ii) Attribute Overview:

Category	Features
1. Identifiers & Schedule	Year, Quarter, Month, DayofMonth, DayOfWeek, FlightDate, Reporting_Airline, Flight_Number_Reporting_Airline
2. Routing	Origin, OriginCityName, OriginState, Dest, DestCityName, DestState
3. Times - Departure	CRSDepTime, DepTime, DepDelay, DepDelayMinutes, DepDel15, TaxiOut, WheelsOff
4. Times - Arrival	WheelsOn, TaxiIn, CRSArrTime, ArrTime, ArrDelay, ArrDelayMinutes, ArrDel15, Cancelled, CancellationCode, Diverted
5. Elapsed Time & Distance	CRSElapsedTime, ActualElapsedTime, AirTime, Distance
6. Delay Cause Breakdown	CarrierDelay, WeatherDelay, NASDelay, SecurityDelay, LateAircraftDelay

(iii) Summary Statistics:

Key Attribute	Description	Valid Count	Mean	Std	Median
DepDelay	Departure delay in minutes	6,903,028	13.56	57.54	-2
ArrDelay	Arrival delay in minutes	6,879,484	8.50	59.74	-6
Distance	Flight Distance in miles	7,001,619	844.45	601.98	692
AirTime	Time spent airborne in minutes	6,879,484	116.84	71.26	100
WeatherDelay	Delay minutes attributed to weather	1,534,638	4.63	34.82	0

Note: Valid Count represents the number of non-missing observations for each attribute. Summary statistics are calculated using valid observations only.

Flight Status Statistics:

Statistics	Count	Percentage
Total Scheduled Flights	7,001,619	100%
Cancelled Flights	102,876	1.47%
Diverted Flights	19,258	0.28%
Flights Arriving 15+ Minutes Late	1,534,638	21.92%
Weather-related Cancellations	63,404	0.91%

Note: All percentages are calculated relative to the total number of scheduled flights.

Dataset 3. Weather 2025: GHCN-Daily

a. Description: NOAA Global Historical Climatology Network Daily (GHCN-Daily). It is NOAA's official quality-controlled daily climate record. We use 2025 data from the weather station at each of 32 major U.S. airports (the FAA large hubs plus OAK), with daily max/min temperature, precipitation, snowfall, snow depth, wind and gusts, and fog/thunder/hail/tornado indicators.

b. Source:
https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily (files at https://www.ncei.noaa.gov/pub/data/ghcn/daily/by_station/)

c. Bulk Download:
BGHCN-Daily publishes one gzipped CSV per station at .../by_station/{STATION_ID}.csv.gz, containing that station's full history. An airport's station ID is "USW000" followed by the station's 5-digit WBAN number, e.g. JFK → USW00094789. Our Python script (https://drive.google.com/file/d/1xkRCsuqxl_dVvLcKe9APrrACtFigoh3I/view?usp=drive_link)

downloads the 32 station files, retrying if a connection drops. It keeps only the 2025 rows, unchanged, and saves one CSV per airport.

d. (i) Relevant Size Statistics: 4.74 MB, 120,371 rows, 9 attributes (airport code, station id, date, element, value, measurement flag, quality flag, source flag, observation time). Covers 2025-01-01 to 2025-12-31, with 23 distinct elements and 3,007 to 4,142 rows per station (about 10 measurements per day)

(ii) Summary statistics:

Statistics	**Unit**	**Count**	**Mean**	**Median**	**Std**
TMAX	tenths °C	11670	219.74	233	101.96
TMIN	tenths °C	11671	120.89	133	97.56
PRCP	tenths mm	11672	21.3	0	71.74
AWND (avg wind)	tenths m/s	11653	37.91	35	16.62

Weather Event Statistics

Statistics	Count	Percentage
Total airport-days	11676	100%
Fog (WT01 or WT02)	3701	31.7%
Days with precipitation (PRCP > 0)	3006	25.75%
Days with snowfall (SNOW > 0)	256	2.19%
Freezing days (TMIN ≤ 0 °C)	1597	13.68%

Note: an airport-day is one airport on one date (32 airports × 2025). All percentages are relative to the total number of airport-days.

Queries:

1. How does the average flight ticket price change as the departure date approaches?
   Calculate average ticket prices across different booking windows, such as 1–7, 8–14, 15–30, and 31–60 days before departure.
2. Which airlines offer the lowest average ticket prices for each route?
   Compare airlines operating on the same routes and rank them based on average prices.
3. Which airlines have the highest cancellation rates and average delays on specific routes?
   Aggregate flight performance records by airline and route to identify historically less reliable options.

4. How do weather conditions affect flight delays and cancellations at major U.S. airports?
   Join flight status records with daily weather observations to compare delay and cancellation rates under different weather conditions.
5. Which flight routes have the greatest variation in ticket prices across different booking windows?
   Calculate the standard deviation of ticket prices and identify routes with the highest price volatility.
6. Which airlines offer the best balance between ticket price and historical flight reliability?
   Combine aggregated pricing and flight performance data to compare airlines based on average ticket prices, cancellation rates, and delays.
7. How do historical weather conditions and cancellation rates vary across seasons for different airports?
   Analyze seasonal patterns using weather observations and flight performance data